

| 5(a)                                                                                                                                                                              | <table><tr><th>description</th><th>letter</th></tr><tr><td>the chamber of the heart that receives blood from the vena cava</td><td>H ;</td></tr><tr><td>the chamber of the heart that pumps blood directly into the pulmonary artery</td><td>G ;</td></tr><tr><td>an artery with a large quantity of elastic fibres and proportionately very little smooth muscle in the tunica media<br/><i>if more than one letter given – all must be correct</i></td><td>A / C / K ;</td></tr><tr><td>a vein transporting oxygenated blood</td><td>B ;</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                            | description | letter | the chamber of the heart that receives blood from the vena cava | H ; | the chamber of the heart that pumps blood directly into the pulmonary artery | G ; | an artery with a large quantity of elastic fibres and proportionately very little smooth muscle in the tunica media<br><i>if more than one letter given – all must be correct</i> | A / C / K ; | a vein transporting oxygenated blood | B ; | 4 |
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| description                                                                                                                                                                       | letter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |        |                                                                 |     |                                                                              |     |                                                                                                                                                                                   |             |                                      |     |   |
| the chamber of the heart that receives blood from the vena cava                                                                                                                   | H ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |        |                                                                 |     |                                                                              |     |                                                                                                                                                                                   |             |                                      |     |   |
| the chamber of the heart that pumps blood directly into the pulmonary artery                                                                                                      | G ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |        |                                                                 |     |                                                                              |     |                                                                                                                                                                                   |             |                                      |     |   |
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| a vein transporting oxygenated blood                                                                                                                                              | B ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |        |                                                                 |     |                                                                              |     |                                                                                                                                                                                   |             |                                      |     |   |
| 5(b)(i)                                                                                                                                                                           | <p>any <b>two</b> from:</p> <p><i>allows, blood / water, to</i></p> <p>resist / AW, changes in temperature ;</p> <p><b>A</b> slow to change when temperatures change</p> <p><b>A</b> prevents short-term temperature fluctuations / AW</p> <p><b>A</b> a lot of energy required to cause a (small) temperature change</p> <p>helps, to maintain a, constant / stable, body temperature</p> <p><b>I</b> optimum temperature</p> <p><b>A</b> <i>ref. to</i> constant / stable, blood temperature for homeostasis</p> <p><b>or</b></p> <p>helps to distribute heat evenly around the body / in thermoregulation ; AW</p> <p>maintain a, steady / constant, external environment for, tissues / cells ;</p> <p>(blood) temperature affects / constant (blood) temperature maintains,</p> <p>solubility of solutes in blood /</p> <p>efficient enzyme functioning AW (in blood / cells) /</p> <p>viscosity / AW, of blood</p> <p>temperature of tissue fluid</p> <p>protein, function / structure / AW ;</p> | 2           |        |                                                                 |     |                                                                              |     |                                                                                                                                                                                   |             |                                      |     |   |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks |
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| 5(b)(ii) | <p>any <b>three</b> from:</p> <ol style="list-style-type: none"> <li>1 solvent for, assimilates / photosynthates / organic substances / mineral ions (present in phloem sap) ;<br/><b>A</b> named substances / ions</li> <li>2 <i>idea that</i> water enters phloem (sieve tube) to counteract lower water potential caused by sucrose ;<br/><b>R</b> lower water potential gradient</li> <li>3 increases, water potential / volume (at source) ; <i>context must be the source</i></li> <li>4 increases / high, hydrostatic pressure (at source) /<br/>hydrostatic pressure gradient (between source and sink) forms ; <b>bod</b> hydrostatic pressure difference created / AW</li> <li>5 water, moves out / exits, (by osmosis) at sink (following solutes), so lowers hydrostatic pressure ;<br/><i>ref. to hydrostatic needed only once</i></li> <li>6 <i>ref. to</i> cohesive nature of water <b>or</b> cohesion / hydrogen bonding, between water molecules, qualified with a correct context ;<br/>e.g. for bulk flow<br/>for movement, of phloem sap / in phloem<br/>for translocation<br/>to maintain, continuous, column / stream (of sap)</li> </ol> | 3     |